

Pavan Vynatheya

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Research Experience

Canadian Institute for Theoretical Astrophysics, University of Toronto

CITA POSTDOCTORAL FELLOW

Toronto, Canada

Sep. 2024 -

Max Planck Institute for Astrophysics

POSTDOCTORAL RESEARCHER

Garching, Germany

May. 2024 - Aug. 2024

Max Planck Institute for Astrophysics

PHD STUDENT (PIs: ADRIAN S. HAMERS, SELMA DE MINK)

Garching, Germany

Sep. 2020 - Apr. 2024

Inter-University Centre for Astronomy and Astrophysics

MASTER THESIS (PI: KANAK SAHA)

Pune, India

Aug. 2019 - May. 2020

University of Western Ontario

MITACS GLOBALINK RESEARCH FELLOWSHIP (PI: PAULINE BARMBY)

London, Canada

May. 2019 - Aug. 2019

Brera Astronomical Observatory

SUMMER RESEARCH INTERNSHIP (PI: TOMASO BELLONI)

Merate, Italy

May. 2018 - Jul. 2018

Aryabhata Research Institute of Observational Sciences

INDIAN ACADEMY OF SCIENCES SUMMER RESEARCH FELLOWSHIP (PI: SAURABH SHARMA)

Nainital, India

May. 2017 - Jul. 2017

Indian Institute of Astrophysics

SUMMER RESEARCH INTERNSHIP (PI: ANNAPURNI SUBRAMANIAM)

Bengaluru, India

Jun. 2016 - Jul. 2016

Education

Max Planck Institute for Astrophysics

PHD IN ASTRONOMY (DEGREE FROM LUDWIG MAXIMILIAN UNIVERSITY OF MUNICH)

Garching, Germany

Sep. 2020 - May. 2024

Part of the International Max Planck Research School (IMPRS) Munich

Overall grade: *magna cum laude*

Indian Institute of Science Education and Research (IISER) Kolkata

BS-MS DUAL DEGREE IN PHYSICS (DST INSPIRE FELLOW)

Mohanpur, India

Major in Physics + Minor in Mathematics

Cumulative Grade Point Average: 9.19 out of 10

Aug. 2015 - Jul. 2020

Publications

FIRST AUTHOR

- Vynatheya, Pavan; Dessart, Luc; Ryu, Taeho; Pakmor, Rüdiger: 2026, arXiv:2601.20133 (Submitted to A&A)
"Asymmetrical thermonuclear supernovae triggered by the tidal disruption of white dwarfs"
- Vynatheya, Pavan; Ryu, Taeho; Wang, Chen; Sills, Alison; Pakmor, Rüdiger: 2026, ApJ, 999, 64
"The collision and merger products of stars do not look alike: A magnetohydrodynamics comparison"
- Vynatheya, Pavan; Ryu, Taeho; Pakmor, Rüdiger; de Mink, Selma E.; Perets, Hagai B.: 2024, A&A, 685, A45
"Simulating the tidal disruption of stars by stellar-mass black holes using moving-mesh hydrodynamics"
- Vynatheya, Pavan; Mardling, Rosemary A.; Hamers, Adrian S.: 2023, MNRAS, 525, 2388
"Quadruple-star systems are not always nested triples: a machine learning approach to dynamical stability"
- Vynatheya, Pavan; Hamers, Adrian S.; Mardling, Rosemary A.; Bellinger, Earl P.: 2022, MNRAS, 516, 4146
"Algebraic and machine learning approach to hierarchical triple-star stability"
- Vynatheya, Pavan; Hamers, Adrian S.: 2021, ApJ, 926, 195
"How important is secular evolution for black hole and neutron star mergers in 2+2 and 3+1 quadruple-star systems?"

CO-AUTHOR

- Preece, Holly P.; Vigna-Gómez, Alejandro; Rajamuthukumar, Abinaya Swaruba; Vynatheya, Pavan; Klencki, Jakub: 2024, arXiv:2412.14022 (Submitted to A&A)
"The Evolution of Massive Stellar Multiplicity in the Field I. Numerical simulations, long-term evolution and final outcome"

3. Britavskiy, Nikolay; Renzo, Mathieu; Nazé, Yael; Rauw, Gregor; **Vynatheya, Pavan**: 2024, A&A, 684, A35
"Tracing the evolution of short-period binaries with super-synchronous fast-rotators"
2. Hamers, Adrian S.; Rantala, Antti; Neunteufel, Patrick; Preece, Holly; **Vynatheya, Pavan**: 2021, MNRAS, 502, 4479
"Multiple Stellar Evolution: a population synthesis algorithm to model the stellar, binary, and dynamical evolution of multiple-star systems"
1. Bogensberger, David; Ponti, Gabriele; Jin, Chichuan; Belloni, Tomaso M.; Pan, Haiwu; Nandra, Kirpal; Russell, Thomas D.; Miller-Jones, James C. A.; Muñoz-Darias, Teo; **Vynatheya, Pavan**; Vincentelli, Federico: 2020, A&A, 641, A101
"An underlying clock in the extreme flip-flop state transitions of the black hole transient Swift J1658.2-4242"

Awards & Fellowships

2024-27	CITA Postdoctoral Fellowship , Postdoctoral research at the Canadian Institute for Theoretical Astrophysics, University of Toronto	Toronto, Canada
2023	Poster Award , 3,2,1: Massive Triples, Binaries and Mergers conference – poster on the dynamical stability of triple- and quadruple-star systems	Leuven, Belgium
2022	Kippenhahn Award , Best PhD student paper in 2021 at the Max Planck Institute for Astrophysics – paper on gravitational wave progenitors in quadruple-star systems	Garching, Germany
2018	Mitacs Globalink Research Internship , Summer research at the University of Western Ontario	London, Canada
2017	Indian Academy of Sciences Summer Research Fellowship , Summer research at the Aryabhata Research Institute of Observational Sciences	Nainital, India
2015-20	Innovation in Science Pursuit for Inspired Research (INSPIRE) Fellowship , Awarded by the Department of Science and Technology to study science at IISER Kolkata for 5 years	Mohanpur, India

Talks & Workshops

INVITED TALKS

Mar. 2026	Astro seminar , Kavli Insitute for the Physics and Mathematics of the Universe	Kashiwa, Japan
Sep. 2025	CITA seminar , Canadian Institute for Theoretical Astrophysics, University of Toronto	Toronto, Canada
Jun. 2025	Seminar on Stellar Astrophysics , Max Planck Institute for Astrophysics	Garching, Germany
Dec. 2024	TASTY seminar , Department of Astronomy and Astrophysics, University of Toronto	Toronto, Canada
Apr. 2024	MPA institute seminar , Max Planck Institute for Astrophysics	Garching, Germany
Mar. 2024	Collisional N-body dynamics workshop , Max Planck Institute for Astrophysics	Garching, Germany
Oct. 2022	Galaxies, Stars & Cosmology group seminar , Marseille Astrophysics Laboratory	Marseille, France (online)

SEMINAR TALKS

Dec. 2025	Astrophysics & Cosmology seminar , Indian Institute of Science	Bengaluru, India
Oct. 2024	Astronomy journal club , Department of Physics and Astronomy, McMaster University	Hamilton, Canada
Oct. 2024	CITA Jamboree , Canadian Institute for Theoretical Astrophysics, University of Toronto	Toronto, Canada
Sep. 2023	Stars and Compact Objects group seminar , Center for Computational Astrophysics, Flatiron Institute	New York, USA
Aug. 2023	Theoretical Astrophysics Including Relativity group seminar , Theoretical Astrophysics Including Relativity and Cosmology, California Institute of Technology	Pasadena, USA
Aug. 2023	Stellar & Galactic Astronomy group seminar , Division of Astronomy & Astrophysics, University of California Los Angeles	Los Angeles, USA
Jun. 2023	Massive stars and stellar populations group seminar , Anton Pannekoek Institute for Astronomy, University of Amsterdam	Amsterdam, Netherlands
Apr. 2023	Seminar on Stellar Astrophysics , Max Planck Institute for Astrophysics	Garching, Germany
Nov. 2022	Astrophysical relativity group seminar , International Centre for Theoretical Sciences	Bengaluru, India
Oct. 2022	Seminar on Stellar Astrophysics , Max Planck Institute for Astrophysics	Garching, Germany
Sep. 2022	Astrophysics whiteboard session , School of Physics & Astronomy, Monash University	Clayton, Australia
Oct. 2021	Seminar on Stellar Astrophysics , Max Planck Institute for Astrophysics	Garching, Germany
Jul. 2021	Max Planck Institute for Astrophysics summer retreat , Schloss Ringberg	Kreuth, Germany

CONTRIBUTED TALKS

May. 2026	IAU Symposium 406: Future Landscape of Astrophysical Transients , Hotel Kakola (organized by University of Turku)	Turku, Finland
Feb. 2026	Multi-Messenger Astrophysics in the Dynamic Universe , Yukawa Institute for Theoretical Physics, Kyoto University	Kyoto, Japan

Jun. 2025	IAU Symposium 398 & MODEST-25: Compact Objects and Binaries in Dense Stellar Systems , Seoul National University	Seoul, South Korea
Apr. 2024	Anticipating the Rising Tide of Tidal Disruption Events: Theory and Observations , Kavli Institute for Theoretical Physics, University of California Santa Barbara	Santa Barbara, USA
Sep. 2023	“Two in a million” – The interplay between binaries and star clusters , European Southern Observatory	Garching, Germany
Aug. 2023	MODEST-23: Star clusters in the post-pandemic era , Northwestern University	Evanston, USA
May. 2022	Growing Black Holes: Accretion and Mergers , Radisson Hotel Kathmandu (organized by Astrophysics and Space Science Observatory of Bologna)	Kathmandu, Nepal
Jan. 2022	Dynamical Formation of Gravitational Wave Sources , Aspen Center for Physics	Aspen, USA

SCHOOLS

Jan. 2024	Nordic Winter School on Multimessenger Astrophysics , Thon Hotel Skeikampen	Skeikampen, Norway
Jul. 2022	Summer School for Astrostatistics at Crete , Department of Physics, University of Crete	Heraklion, Greece
Aug. 2021	NBIA Summer School on Gravitational Wave Astrophysics , Niels Bohr International Academy, University of Copenhagen	Copenhagen, Denmark
Dec. 2018	IIST Astronomy and Astrophysics School , Indian Institute of Space Science and Technology	Thiruvananthapuram, India
Feb. 2017	Telescope-making workshop , Indian Institute of Science Education and Research Kolkata	Mohanpur, India
Dec. 2015	Vijyoshi Science Camp , Eastern Zonal Cultural Centre (organized by Indian Institute of Science Education and Research Kolkata)	Kolkata, India

CONFERENCE POSTERS

Jul. 2023	3,2,1: Massive Triples, Binaries and Mergers , Katholieke Universiteit Leuven	Leuven, Belgium
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Skills

Astrophysics code	Moving-mesh hydrodynamics: AREPO , Multiple-star population synthesis: MSE , <i>N</i> -body: MSTAR , REBOUND , Detailed stellar evolution: MESA
Programming	python (advanced + machine learning), C/C++ (intermediate), Java (basic)
Web design	HTML (advanced), CSS (advanced), jQuery/Javascript (intermediate), PHP (basic)
Languages	English (fluent), Kannada (native), Tamil (native), Hindi (fluent)

Students

2024-25	Gurman Sachdeva , Undergraduate student, Department of Astronomy and Astrophysics, University of Toronto	Toronto, Canada
2025-26	Yung-Hsin (Stephanie) Lee , Post-undergraduate student, Department of Astronomy and Astrophysics, University of Toronto	Toronto, Canada

Activities

2025-	Referee , Astrophysical Journal (ApJ), Publications of the Astronomical Society of Japan (PASJ)	
2021-23	Student representative , International Max Planck Research School (IMPRS) Munich	Garching, Germany
2021-22	Co-organizer , Seminar on Stellar Astrophysics (SESTAS)	Garching, Germany
2018 and 2019	Web designer , IISER Kolkata team at the International Genetically Engineered Machine (iGEM) competition - Gold medal winners	Boston, USA (did not attend)
2016-17	Web designer , Inquivesta - Science fest of IISER Kolkata	Mohanpur, India
2016-17	Convener , IISER Kolkata Music club	Mohanpur, India